

Assessment -2:

Commands:

```
docker run -d -p 8080:8080 --name airflow apache/airflow standalone
```

```
docker ps - here you will see the port number
```

```
docker logs airflow
```

```
docker exec -it airflow bash
```

```
cd dags
```

```
cat > daily_attendance_dag.py
```

Code:

```
from airflow import DAG
```

```
from airflow.operators.python import PythonOperator
```

```
from datetime import datetime
```

```
ATTENDANCE_FILE="/tmp/attendance.txt"
```

```
ATTENDANCE_REPORT="/tmp/attendance_report.txt"
```

```
def create_attendance_file():
```

```
    with open(ATTENDANCE_FILE,"w") as file:
```

```
        file.write("Aarav,Present\nPriya,Present\nRahul,Absent\nSneha,Present\nKiran,Absent\nAnanya,Present\nVikram,Present\nMeera,Absent\nFarhan,Present\nDivya,Present\n")
```

```
def read_attendance_file():
```

```
    with open(ATTENDANCE_FILE,"r") as file:
```

```
        data=file.readlines()
```

```
    for line in data:
```

```
        print(line.strip())
```

```
def count_total_students(ti):
```

```
    with open(ATTENDANCE_FILE,"r") as file:
```

```
        data=file.readlines()
```

```
    total_students=len(data)
```

```
    print("Total Students =",total_students)
```

```
ti.xcom_push(key="total_students",value=total_students)
```

```
def count_present_students(ti):  
    with open(ATTENDANCE_FILE,"r") as file:  
        data=file.readlines()  
    present_count=0  
    for line in data:  
        name,status=line.strip().split(",")  
        if status=="Present":  
            present_count+=1  
    print("Present Students =",present_count)  
    ti.xcom_push(key="present_students",value=present_count)
```

```
def count_absent_students(ti):  
    with open(ATTENDANCE_FILE,"r") as file:  
        data=file.readlines()  
    absent_count=0  
    for line in data:  
        name,status=line.strip().split(",")  
        if status=="Absent":  
            absent_count+=1  
    print("Absent Students =",absent_count)  
    ti.xcom_push(key="absent_students",value=absent_count)
```

```
def calculate_attendance_percentage(ti):  
    total_students=ti.xcom_pull(task_ids="count_total_students",key="total_students")  
  
    present_students=ti.xcom_pull(task_ids="count_present_students",key="present_students")  
  
    attendance_percentage=(present_students/total_students)*100  
    print("Attendance Percentage =",str(int(attendance_percentage))+"%")  
    ti.xcom_push(key="attendance_percentage",value=int(attendance_percentage))
```

```
def list_absent_students():
```

```

with open(ATTENDANCE_FILE,"r") as file:
    data=file.readlines()
print("Absent Students List")
for line in data:
    name,status=line.strip().split(",")
    if status=="Absent":
        print(name)

def generate_attendance_report(ti):
    total_students=ti.xcom_pull(task_ids="count_total_students",key="total_students")

    present_students=ti.xcom_pull(task_ids="count_present_students",key="present_students")

    absent_students=ti.xcom_pull(task_ids="count_absent_students",key="absent_students")

    attendance_percentage=ti.xcom_pull(task_ids="calculate_attendance_percentage",key="attendance_percentage")

    if attendance_percentage>=75:
        status="Good"
    else:
        status="Needs Improvement"

    with open(ATTENDANCE_REPORT,"w") as file:
        file.write(f"Daily Attendance Report\nTotal Students = {total_students}\nPresent Students = {present_students}\nAbsent Students = {absent_students}\nAttendance Percentage = {attendance_percentage}%\nStatus = {status}")

default_args={
    "owner":"airflow"
}

```

```
with DAG(
    dag_id="daily_attendance_processor",
    default_args=default_args,
    start_date=datetime(2024,1,1),
    schedule=None,
    catchup=False
) as dag:

    create_attendance_file_task=PythonOperator(
        task_id="create_attendance_file",
        python_callable=create_attendance_file
    )

    read_attendance_file_task=PythonOperator(
        task_id="read_attendance_file",
        python_callable=read_attendance_file
    )

    count_total_students_task=PythonOperator(
        task_id="count_total_students",
        python_callable=count_total_students
    )

    count_present_students_task=PythonOperator(
        task_id="count_present_students",
        python_callable=count_present_students
    )

    count_absent_students_task=PythonOperator(
        task_id="count_absent_students",
        python_callable=count_absent_students
    )

    calculate_attendance_percentage_task=PythonOperator(
```

```

task_id="calculate_attendance_percentage",
python_callable=calculate_attendance_percentage
)

```

```

list_absent_students_task=PythonOperator(
    task_id="list_absent_students",
    python_callable=list_absent_students
)

```

```

generate_attendance_report_task=PythonOperator(
    task_id="generate_attendance_report",
    python_callable=generate_attendance_report
)

```

```

create_attendance_file_task>>read_attendance_file_task>>count_total_students_task>>count_present_students_task>>count_absent_students_task>>calculate_attendance_percentage_task>>list_absent_students_task>>generate_attendance_report_task

```

```

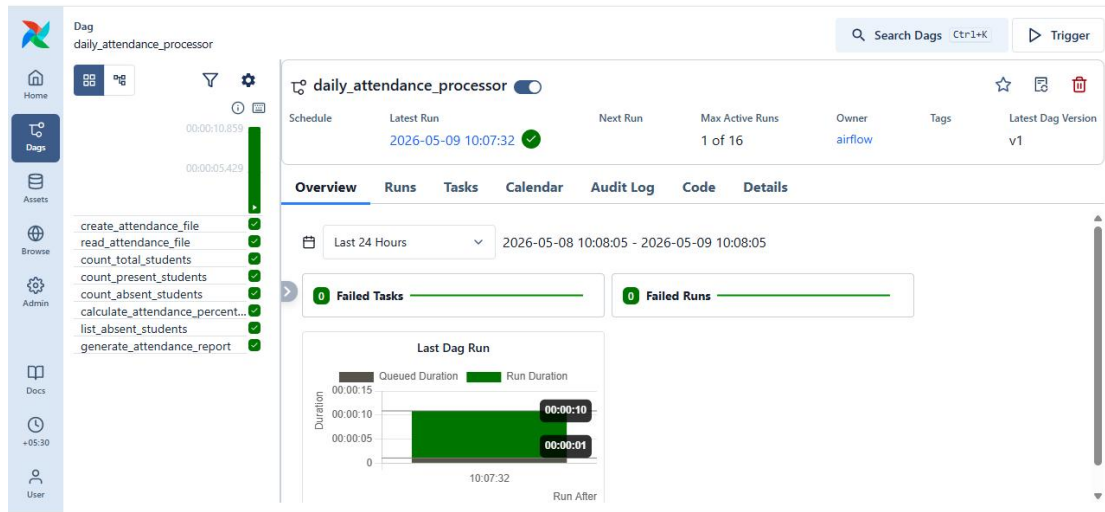
airflow@db6fbd791f6:/opt/airflow/dags$ ls
daily_attendance_dag.py

```

Output Screenshots:

The screenshot displays the Apache Airflow web interface. The left sidebar contains navigation links for Home, Dags, Assets, Browse, Admin, Docs, and User. The main content area is titled 'Dags' and shows a search bar and filters for 'All', 'Failed', 'Queued', 'Running', 'Success', and 'Required Actions'. Below the filters, a table lists the DAGs. The first DAG is 'daily_attendance_processor', which is shown with its latest run status as 'Success' on '2026-05-09 10:07:32'.

Schedule	Latest Run	Next Run
	2026-05-09 10:07:32 ✔	



```
airflow@db6fbdc791f6:/opt/airflow/dags$ cat /tmp/attendance.txt
Aarav,Present
Priya,Present
Rahul,Absent
Sneha,Present
Kiran,Absent
Ananya,Present
Vikram,Present
Meera,Absent
Farhan,Present
```

```
Daily Attendance Report
Total Students = 10
Present Students = 7
Absent Students = 3
Attendance Percentage = 70%
Status = Needs Improvementairflow@db6fbdc791f6
```